

Guide: A Pair of Parity Words That Forces the Trivial Cycle

Collatz proof project

September 5, 2026

The result

Take any finite binary word u . Put an odd step at its beginning and an even step at its end, making $1u0$. Now swap only those endpoint bits, making $0u1$.

Lean proves that these two words cannot both describe complete returns from natural Collatz seeds, unless u is empty. In the empty case they are simply 10 and 01, starting at 1 and 2 in the usual cycle. The returning seeds do not even have to lie in the same orbit.

Why this works

Every parity word gives an exact affine equation for its endpoint. For a cycle, this expresses its starting seed as a correction divided by $D = 2^{\text{length}} - 3^{\text{odd count}}$. On a positive natural cycle, D is positive and odd.

For our paired words, a short cancellation shows that D must divide a power of two. Therefore $D = 1$. If the middle word is nonempty, reducing $2^{\text{length}} = 3^{\text{odd count}} + 1$ modulo 8 gives a contradiction. Only the empty middle survives, and its return equations give seeds 1 and 2.

How this helps the research

The nested Knight formalization uses this cancellation to exclude a class of evenly spaced parity patterns called high cycles. The new theorem puts the obstruction directly into this project's actual-orbit language and does not require a palindrome condition on the middle word.

The next task is to prove that every primitive rational mechanical pattern contains the required pair as rotations of one period. Exact finite checks confirm this for all reduced slopes with denominator up to 80. That finite check does not prove the universal statement. Nonprimitive periods and eventual tails also need explicit treatment.

The limitation

Arbitrary parity words need not have the required pair. The word 11100 is a concrete example with no such pair among its rotations. Its associated cyclic value is the fraction $19/5$, not an integer counterexample.

This result is a formal integration of a known cancellation idea, not a proof of Collatz or a claim of mathematical novelty. The full conjecture still requires general descent, or both escape and nontrivial-cycle exclusion.

Verification

From `lean/`, run `lake build Collatz.EndpointCycles`. From the root, with Lean on the executable path, run `python3 analysis/audit_theorems.py --build`. The three exact finite controls run with `python3 -m unittest discover -s analysis -p test_endpoint_cycles.py`. The companion technical paper gives the algebra, exact theorem scope, provenance, and remaining formal bridge.